



# TARGET-90

**JKBOSE CLASS-11<sup>TH</sup>**

**PROBABILITY**

One Shot

**MATHS**

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# Topics To Be Covered

- 1 ... Introduction To Probability
- 2 ... Terms Related to Probability
- 3 ... Important Question + PYQ's.



# PROBABILITY



## What is Probability?

Probability is a measure of **how likely** an event is to happen.

## Range of Probability



## Formula (for equally likely outcomes)

$$P(\text{event}) = \frac{\text{Number of favorable outcomes}}{\text{Total number of possible outcomes}}$$

## Example 1: Rolling a Die

Probability of rolling a 4

$$P(4) = \frac{1}{6} \approx 0.167$$



## Example 2: Even Number

Even numbers = 2, 4, 6

$$P(\text{even}) = \frac{3}{6} = \frac{1}{2} = 50\%$$



## Key Points

- Probability always lies between 0 and 1.
- The sum of probabilities of all outcomes is 1.
- More probability means a higher chance of an event happening.

## Real Life Example



In a fair coin toss,

$$P(\text{Head}) = \frac{1}{2}$$

$$P(\text{Tail}) = \frac{1}{2}$$

## Visual Representation



A probability can also be shown using diagrams (like Venn diagrams) for combined events.

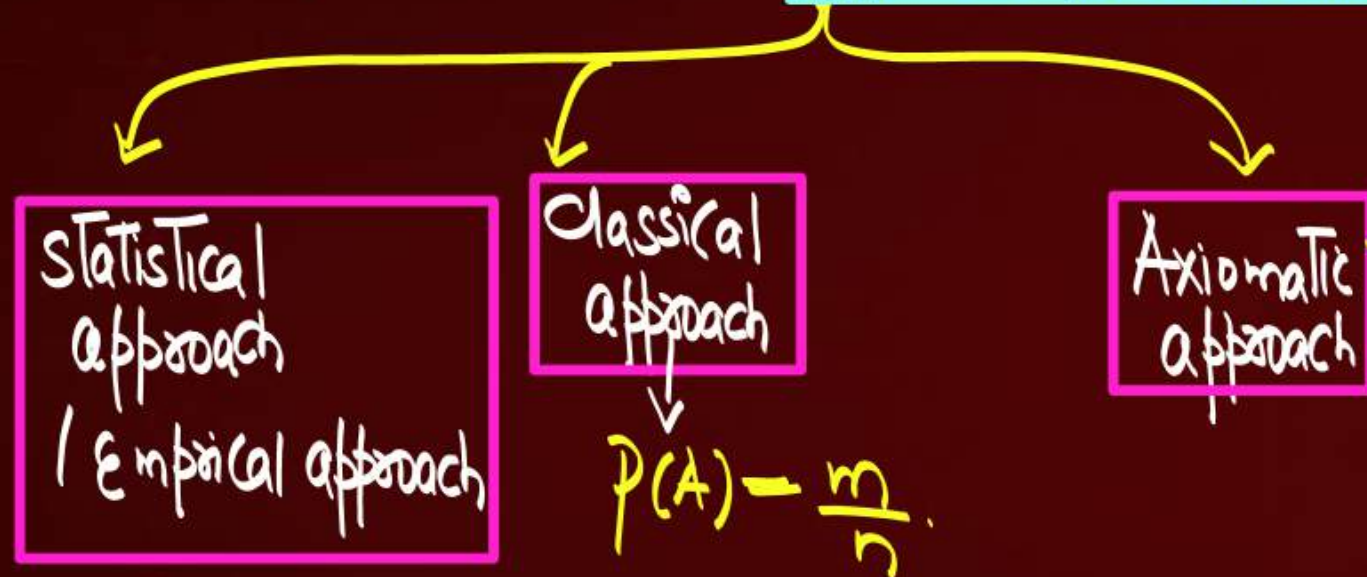


# INTRODUCTION



**Definition:** Probability is the measure of uncertainty of various phenomenon's.

## Approaches for Measurement



$$P(A) = \frac{m}{n}$$

**Experiment:** An operation which can produce some well defined outcomes is known as an experiment.

## Random experiment (Probabilistic experiment)

(Outcome cannot be predicted in advance)

\* **Sample space:** The set of all possible outcomes of a Random experiment is known as the Sample space.

\* **Event:** Any Subset of the Sample space 'S' is known as an event.

## Deterministic experiment

(Outcome can be predicted in advance)



## QUESTION

A bag contains 4 identical red balls and 3 identical black balls. The experiment consists of drawing one ball, then putting it into the bag and again drawing a ball. Then, the possible outcomes of this experiment is.

$$S = \{RR, RB, BR, BB\}$$

- 1 {RR, BB} ✗
- 2 {RR, BB, RR} ✗
- 3 {BB, R} ✗
- 4 None of these ✓



## QUESTION



Pair of dice is rolled Describe the following events associated with this random experiment

A = Getting the sum of the number on the dice is 2.

B = Getting the sum of the number is multiple of 11.

C = Getting the sum of the number is multiple of 12.

Which of these events are compound events?

- 1 ~~A~~
  - 2 B ✓
  - 3 ~~C~~
  - 4 All of these ✓
- $A = \{(1,1)\} \rightarrow$  Simple event/elementary event
- $B = \{(5,6), (6,5)\} \rightarrow$  Compound event
- $C = \{(6,6)\} \rightarrow$  Simple/elementary event.



## EXAMPLE



Consider the experiment of rolling a die. Let A be the event 'getting a prime number', B be the event 'getting an odd number'. Write the sets representing the events

(i) A or B

(ii) A and B

(iii) A but not B

(iv) 'not A'.

(Sol) Here  $S = \{1, 2, 3, 4, 5, 6\}$ .

$$A = \{2, 3, 5\}, B = \{1, 3, 5\}$$

(i)  $A \text{ or } B = A \cup B = \{1, 2, 3, 5\}$

(ii)  $A \text{ and } B = A \cap B = \{3, 5\}$

(iii)  $A \text{ but not } B = A - B = \{2\}$

(iv)  $\text{not } A = A' = \{1, 4, 6\}$

## QUESTION

Let  $S = \{1, 2, 3, 4, 5, 6\}$  and  $E = \{1, 3, 5\}$ , then  $E'$  is

- 1  $\{2, 4\}$  ✗
- 2  $\{3, 6\}$  ✗
- 3  $\{1, 2, 4\}$  ✗
- 4  $\{2, 4, 6\}$  ✓

$$E' = \{2, 4, 6\}$$



## QUESTION



A pair of dice is rolled, if the outcomes is a doublet, a coin is tossed. Then, the total number of outcomes for this experiment is



1 40 ✗

2 41 ✗

3 42 ✓

4 43 ✗

$$30 + 6 \times 2 = 42.$$

Non-Doublet → Doublet



## QUESTION



A die is rolled. Let  $E$  be the event "die shows 4" and  $F$  be the event "die shows even number". Are  $E$  and  $F$  mutually exclusive?  $(E \cap F = \phi) \rightarrow E \text{ \& F are Mutually exclusive.}$

(Sol) Here  $E = \{4\}$  and  $F = \{2, 4, 6\}$ .

$$\Rightarrow E \cap F \neq \phi$$

$\Rightarrow E \text{ \& F are not Mutually exclusive.}$



## QUESTION



An experiment involves rolling a pair of dice and recording the numbers that come up.  
Describe the following events:

- A: The sum is greater than 8, *H/W*
- B: 2 occurs on either die
- C: The sum is at least 7 and a multiple of 3.

Which pairs of these events are mutually exclusive?



## QUESTION



Three coins are tossed once. Let A denote the event 'three heads show', B denote the event "two heads and one tail show", C denote the event "three tails show" and D denote the event 'a head shows on the first coin'. Which events are

(i) Mutually exclusive?  <sup>$A \& B; A \& C; B \& C; C \& D$</sup>

(ii) Simple?  $(A, C)$

(iii) Compound?  
 $(B \& D)$

$$S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$$

$$A = \{HHH\}$$

$$B = \{HHT, HTH, THH\}$$

$$C = \{TTT\}$$

$$D = \{HHH, HHT, HTH, HTT\}$$



# Playing Cards

52 Cards

26 Red Cards

26 Black Cards

**Hearts**  
(13 cards)

- King
- Queen
- \* Jack
- Ace
- 2-10

**Diamonds**  
(13 cards)

**Clubs**  
(13 cards)

**Spades**  
(13 cards)



Total no. of kings = 4.  
Total no. of Queens = 4.  
Total no. of Jacks = 4.  
Total no. of face cards = 12.  
Total no. of Honour cards = 16.

\* King + Queen + Jack → Face Cards  
\* Face Cards + Ace → Honour Cards.



# Classical Approach of Probability

In Classical Approach, the probability of an event is defined as the ratio of the no. of cases favourable to the event to the total no. of possible cases. (outcomes are equally likely).

$$* P(A) = \frac{\text{no. of cases favourable to A}}{\text{Total no. of possible outcomes}} = \frac{m}{n}$$

$$\Rightarrow P(A) = \frac{m}{n}$$

$$* 0 \leq P(A) \leq 1$$

\*  $P(A) = 0 \rightarrow A$  is known as an Impossible event.

\*  $P(A) = 1 \rightarrow A$  is known as a Sure event.

$$* S = \{1, 2, 3, 4, 5, 6\}$$

A: Getting a Prime number.

$$\Rightarrow A = \{2, 3, 5\}$$

$$\text{So, } P(A) = \frac{3}{6}$$

$$\Rightarrow \boxed{P(A) = \frac{1}{2}}$$



## EXAMPLE



Card is drawn from a well shuffled deck of 52 cards. If each outcome is equally likely, calculate the probability that the card will be

- (i) A diamond  $\frac{13}{52} = \frac{1}{4}$
- (ii) An not ace  $\frac{48}{52} = \frac{12}{13}$
- (iii) A black card (i.e., a club or, a spade)  $\frac{26}{52} = \frac{1}{2}$
- (iv) Not a diamond  $\frac{39}{52} = \frac{3}{4}$
- (v) Not a black card.  $\frac{26}{52} = \frac{1}{2}$



## EXAMPLE

$$\frac{4!}{2!2!} = \frac{4 \times 3 \times 2 \times 1}{2 \times 2}$$

A committee of two persons is selected from two men and two women. What is the probability that the committee will have (a) no man? (b) one man? (c) two men?

$$(a) \text{ Required Probability} = \frac{{}^2C_0 \times {}^2C_2}{{}^4C_2} = \frac{1}{6}$$

$$(b) \text{ Required Probability} = \frac{{}^2C_1 \times {}^2C_1}{{}^4C_2} = \frac{2 \times 2}{6} = \frac{2}{3}$$

$$(c) \text{ Required Probability} = \frac{{}^2C_2 \times {}^2C_0}{{}^4C_2} = \frac{1}{6}$$

$M_1, M_2, W_1, W_2$

$M_1, M_2 \quad | \quad M_2, M_1$

$M_1, W_1 \quad | \quad W_1, M_2$



## QUESTION



Seven persons are to be seated in a row. The probability that two particular persons sit next to each other is.

$$3! = 3 \times 2 \times 1$$

$$5! = 5 \times 4 \times 3 \times 2 \times 1$$

1  $\frac{1}{3}$  ✗

2  $\frac{1}{6}$  ✗

3  $\frac{2}{7}$  ✓

4  $\frac{1}{2}$  ✗

$$\text{Required Probability} = \frac{6! \times 2!}{7!} = \frac{\cancel{6!} \times 2!}{7 \times \cancel{6!}} \quad \boxed{P_1 P_2 P_3 P_4 P_5 P_6 P_7}$$
$$= \frac{2}{7}$$

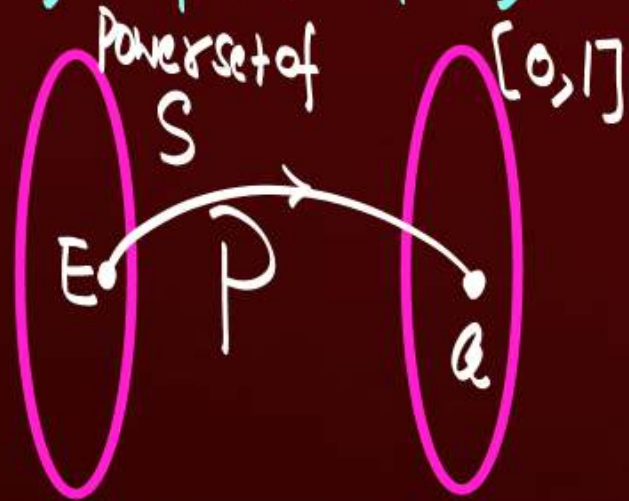


# Axiomatic Approach to Probability

★ **Definition:** Suppose that  $S$  be the Sample space of a Random experiment. The Probability is a function whose domain is the Power set of  $S$  and range is  $[0,1]$  which satisfies the following axioms:

(1)  $P(S) = 1$ .

(2) If  $E$  and  $F$  are mutually exclusive i.e.  $E \cap F = \phi$  then  $P(E \cup F) = P(E) + P(F)$ .



\* We know that  $E \cap \phi = \phi$  for any event  $E$ .

$$\Rightarrow P(E \cup \phi) = P(E) + P(\phi)$$

$$\Rightarrow \cancel{P(E)} = \cancel{P(E)} + P(\phi)$$

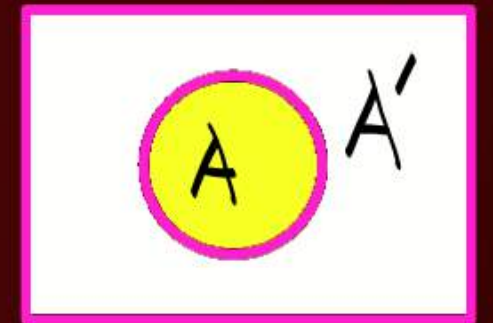
$$\Rightarrow P(\phi) = 0.$$

\* We know that  $E \cup E' = S$  &  $E \cap E' = \phi$ .

$$\Rightarrow P(E \cup E') = P(S) = 1.$$

$$\Rightarrow P(E) + P(E') = 1.$$

$$\Rightarrow P(E') = 1 - P(E)$$





★ Probability of A or B: Clearly from the Venn Diagram, we have;

$$A = (A-B) \cup (A \cap B) \text{ \& } (A-B) \cap (A \cap B) = \phi$$

$$\Rightarrow P(A) = P[(A-B) \cup (A \cap B)] = P(A-B) + P(A \cap B) \quad \text{using ① in ②, we get;}$$

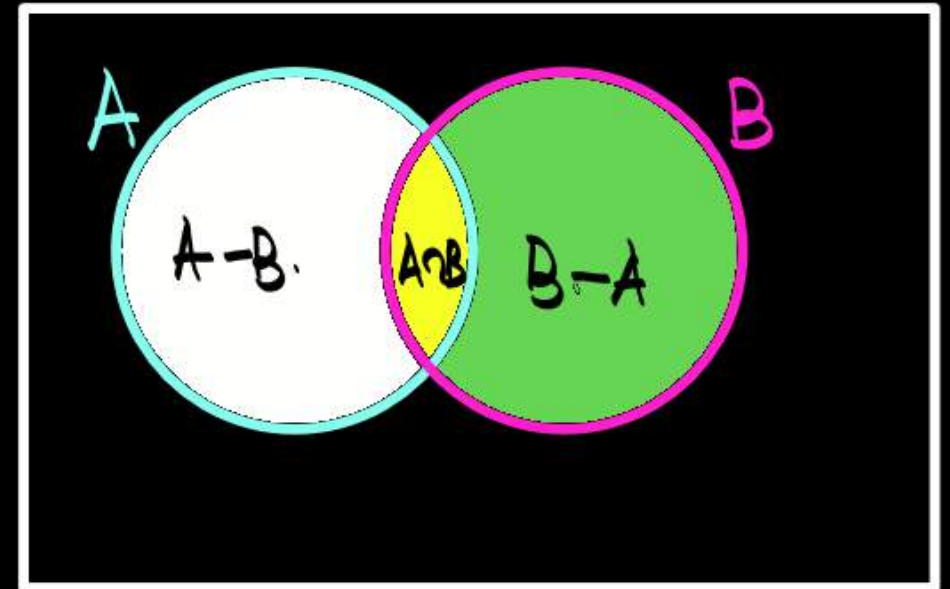
$$\Rightarrow P(A-B) = P(A) - P(A \cap B) \quad \text{--- ①}$$

Again from the Venn-Diagram;

$$A \cup B = (A-B) \cup B \text{ \& } (A-B) \cap B = \phi$$

$$\Rightarrow P(A \cup B) = P[(A-B) \cup B]$$

$$\Rightarrow P(A \cup B) = P(A-B) + P(B) \quad \text{--- ②}$$



$$P(A \cup B) = P(A) - P(A \cap B) + P(B)$$

$$\Rightarrow P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Hence the result.



## EXAMPLE



Let a sample space be  $S = \{\omega_1, \omega_2, \dots, \omega_6\}$ . Which of the following assignments of probabilities to each outcome are valid?

| Outcomes | $\omega_1$     | $\omega_2$     | $\omega_3$    | $\omega_4$    | $\omega_5$     | $\omega_5$ <sup><math>\frac{P(\omega_i)}{n}</math></sup> |
|----------|----------------|----------------|---------------|---------------|----------------|----------------------------------------------------------|
| (a)      | $\frac{1}{6}$  | $\frac{1}{6}$  | $\frac{1}{6}$ | $\frac{1}{6}$ | $\frac{1}{6}$  | $\frac{1}{6}$                                            |
| (b)      | 1              | 0              | 0             | 0             | 0              | 0                                                        |
| (c)      | $\frac{1}{8}$  | $\frac{2}{3}$  | $\frac{1}{3}$ | $\frac{1}{3}$ | $-\frac{1}{4}$ | $-\frac{1}{3}$                                           |
| (d)      | $\frac{1}{12}$ | $\frac{1}{12}$ | $\frac{1}{6}$ | $\frac{1}{6}$ | $\frac{1}{6}$  | $\frac{3}{2}$                                            |
| (e)      | 0.1            | 0.2            | 0.3           | 0.4           | 0.5            | 0.6                                                      |



## QUESTION



What is the Probability that a number selected from the numbers 1, 2, 3, 4, ..., 25, is prime number, when each of the given numbers is equally likely to be selected?

2, 3, 5, 7, 11, 13, 17, 19, 23,

$$\frac{9}{25}$$

1  $\frac{12}{25}$  ✗

2  $\frac{11}{25}$  ✗

3  $\frac{10}{25}$  ✗

4  $\frac{9}{25}$  ✓

## QUESTION



A coin is tossed twice, what is the probability that atleast one tail occurs?

H/W



## QUESTION



A die is thrown, find the probability of following events:

- (i) A prime number will appear,
- (ii) A number greater than or equal to 3 will appear,
- (iii) A number less than or equal to one will appear,
- (iv) A number more than 6 will appear,
- (v) A number less than 6 will appear.

H/w

## QUESTION



A card is selected from a pack of 52 cards.

(a) How many points are there in the sample space? (52)

(b) Calculate the probability that the card is an ace of spades.  $\frac{1}{52}$

(c) Calculate the probability that the card is (i) an ace (ii) black card.

$$\frac{4}{52} = \frac{1}{13}$$

$$\frac{26}{52} = \frac{1}{2}$$



## QUESTION



There are four men and six women on the city council. If one council member is selected for a committee at random, how likely is it that it is a woman?

Imp

## QUESTION



A fair coin is tossed four times, and a person win Re 1 for each head and lose Rs 1.50 for each tail that turns up.

From the sample space calculate how many different amounts of money you can have after four tosses and the probability of having each of these amounts.

$P_i/W_i$



## QUESTION



A letter is chosen at random from the word **'ASSASSINATION'**. Find the probability that letter is (i) a vowel (ii) a consonant

$$\frac{6}{13}$$

$$\frac{7}{13}$$

A A I A I O  
S S S S N T N

## QUESTION



Find in the blanks in following table:

|       | $P(A)$        | $P(B)$        | $P(A \cap B)$  | $P(A \cup B)$ |
|-------|---------------|---------------|----------------|---------------|
| (i)   | $\frac{1}{3}$ | $\frac{1}{5}$ | $\frac{1}{15}$ | ...           |
| (ii)  | 0.35          | ...           | 0.25           | 0.6           |
| (iii) | 0.5           | 0.35          | ...            | 0.7           |



## QUESTION



(PYQ JKBOSE)

If E and F are events such that  $P(E) = \frac{1}{4}$ ,  $P(F) = \frac{1}{2}$ , and  $P(E \text{ and } F) = \frac{1}{8}$ , find

- (i)  $P(E \text{ or } F)$ ,
- (ii)  $P(\text{not } E \text{ and not } F)$ .

(Sol) (i)  $P(E \text{ or } F) = P(E \cup F) = P(E) + P(F) - P(E \cap F)$

$$\Rightarrow P(E \text{ or } F) = \frac{1}{4} + \frac{1}{2} - \frac{1}{8}$$

$$\Rightarrow P(E \text{ or } F) = \frac{2+4}{8} - \frac{1}{8} = \frac{5}{8}$$

(ii)  $P(\text{not } E \text{ and not } F) = P(E' \cap F')$   
 $= P[(E \cup F)'] = 1 - P(E \cup F)$   
 $= 1 - \frac{5}{8} = \frac{3}{8}$  Hence the result.

## QUESTION



Events  $E$  and  $F$  are such that  $P(\text{not } E \text{ or not } F) = 0.25$  State whether  $E$  and  $F$  are mutually exclusive.

H/W



## QUESTION



A and B are events such that  $P(A) = 0.42$ ,  $P(B) = 0.48$  and  $P(A \text{ and } B) = 0.16$ .  
Determine (i)  $P(\text{not } A)$ , (ii)  $P(\text{not } B)$  and (iii)  $P(A \text{ or } B)$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(E') = 1 - P(E)$$

## QUESTION

(PYQ JKPSE)

In Class XI of a school 40% of the students study Mathematics and 30% study Biology. 10% of the class study both Mathematics and Biology. If a student is selected at random from the class, find the probability that he will be studying Mathematics or Biology.

(Sol) Define the events as follows:

M: Student studying Mathematics

B: Student studying Biology.

$$\Rightarrow P(M) = \frac{40}{100}, P(B) = \frac{30}{100}$$

$$\& P(M \cap B) = \frac{10}{100}$$

$$\text{Therefore Required Probability} = P(M \cup B) = P(M) + P(B) - P(M \cap B)$$

$$\Rightarrow \text{Required Probability} = \frac{40}{100} + \frac{30}{100} - \frac{10}{100}$$

$$= \frac{60}{100} = 0.6$$

Hence the result.



## QUESTION



In an entrance test that is graded on the basis of two examinations, the probability of a randomly chosen student passing the first examination is 0.8 and the probability of passing the second examination is 0.7. The probability of passing atleast one of them is 0.95. What is the probability of passing both?

Imp.

H/W.



## QUESTION



(Important)

The probability that a student will pass the final examination in both English and Hindi is 0.5 and the probability of passing neither is 0.1. If the probability of passing the English examination is 0.75, what is the probability of passing the Hindi examination?

(Sol) Define the events as follows:

$E$ : Student passing in English examination.

$H$ : Student passing in Hindi examination.

$$\Rightarrow P(E \cap H) = 0.5, P(E' \cap H') = 0.1.$$

$$\& P(E) = 0.75.$$

$$\text{Here } P(E' \cap H') = 0.1$$

$$\Rightarrow P[(E \cup H)'] = 0.1$$

$$\Rightarrow 1 - P(E \cup H) = 0.1$$

$$\Rightarrow P(E \cup H) = 1 - 0.1$$

$$\Rightarrow P(E) + P(H) - P(E \cap H) = 0.9$$

$$\Rightarrow 0.75 + P(H) - 0.5 = 0.9$$

$$\Rightarrow P(H) + 0.25 = 0.9$$



$$\Rightarrow P(H) = 0.9 - 0.25.$$

$$\Rightarrow P(H) = 0.65.$$

$$\Rightarrow \text{Required Probability} = 0.65.$$

Hence the result.

## QUESTION



In a class of 60 students, 30 opted for NCC, 32 opted for NSS and 24 opted for both NCC and NSS. If one of these students is selected at random, find the probability that

- (i) The student opted for NCC or NSS.
- (ii) The student has opted neither NCC nor NSS. (Important)
- (iii) The student has opted NSS but not NCC.



$$(A \cup B)' = A' \cap B'$$

$$(A \cap B)' = A' \cup B'$$

**Thank**  
*You*

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